

Recording your voice for CHOIR

Everything you need to do, in order. Roughly 90 minutes end to end, of which about 15 minutes is actual talking.

What this produces

A recording good enough to train a synthetic voice that sounds recognisably like you. Your timbre, your accent, your way of speaking, running entirely on-device with no network.

Recording quality is the ceiling on the whole project. The models are commodity now; the data is not. A good microphone in a closet beats a mediocre one in a studio.

Do this first: the 30-second test

Do not record all 75 sentences today. Record three, send them, and find out whether your room and microphone are fighting you. A problem found in 30 seconds costs 30 seconds. The same problem found after the full read costs the whole session, because a recording cannot be repeated under the same conditions.

What you need

Microphone	USB condenser	The single biggest quality factor. A laptop mic works for the test but will not make a sellable voice. Never a headset or earbuds: they compress and denoise, and the model learns the artefacts.
Room	Small and soft	A closet with hanging clothes is genuinely better than a large room. Avoid kitchens and bathrooms. Clothes absorb reflections.
Quiet	No fans, no fridge	Turn off air conditioning, close windows, silence your phone and notifications. A fridge compressor cycling mid-take ruins it.
Software	Audacity (free)	QuickTime cannot produce 24-bit WAV. GarageBand fights you on export. Audacity does exactly what is needed.
Water	Room temperature	Not cold, not fizzy. Dry mouth is audible and gets worse over an hour.

Setting up Audacity

These settings matter. Getting them wrong means re-recording.

1. Open **Audio Setup** and choose your microphone as the recording device.

Not your built-in mic, if you have plugged something in. Audacity does not always switch automatically.

2. Set **Recording Channels** to **1 (Mono)**.

CHOIR is mono end to end. A stereo file from one mic is just the same signal twice and doubles your file size for nothing.

3. Set the **Project Rate** at the bottom-left to **48000 Hz**.

This matches CHOIR's audio format exactly, so nothing has to be resampled before training.

4. Set the quality to **24-bit**.

Audacity > Settings > Quality > Default Sample Format. Gives headroom for later processing. The engine outputs 16-bit; you record with room to spare.

5. Turn **every effect off**.

No compression, no noise reduction, no EQ, no de-essing. CHOIR has its own mastering chain, and processing baked into a recording cannot be undone. Record flat, even though flat sounds worse on its own.

Setting your level

Speak a line at the volume you intend to use for the whole session and watch the recording meter. Aim for the loudest peaks to land around -6 dB. Then do not touch the gain knob again, in this session or any later one. Consistent level across every file matters more than perfect level in any one file.

Too quiet is recoverable. Clipping is not.

If the meter hits the top and turns red, the waveform is squared off and that information is gone forever. No processing recovers it. When in doubt, record quieter.

How to read

The way you read is what the model learns. It copies your habits, including the ones you do not notice.

- **Read to one person.** Imagine explaining something to a single attentive friend across a table. Not announcing, not performing, not reading a bedtime story.
- **Keep your pace even.** Not slow and careful. Your natural speaking rate, held steady across all 75 lines.
- **Stay still.** Same distance from the mic throughout, about 20 cm, slightly off to one side so your breath does not hit the capsule directly. Moving closer and further teaches the model a wobble.
- **Do not act the punctuation.** Read questions as questions and commands as commands, but naturally. Exaggerated intonation is learned and reproduced on every sentence afterwards.
- **Leave the mistakes in the file.** Do not stop the recording. Pause about one second, then say the whole line again from the beginning.

The recording session

1. Record 10 seconds of silence.

Press record, sit completely still, say nothing, stop. This is your room tone. It documents your noise floor and proves the room did not change between sessions.

2. Press record once, and read all 75 lines.

One continuous take. Do not stop and start between sentences. Splitting one long file is mechanical and I can do it; managing 75 separate files by hand is miserable and error-prone.

3. Pause about one second between sentences.

Sit still and silent during the pause. Those gaps are how the file gets split automatically, so they need to be genuinely quiet. Do not shuffle paper or click a mouse in the gap.

4. Take a break if you need one.

Stop the recording, rest, then start a second file. Do not push through fatigue: a tired voice has a different timbre and the model will learn the tiredness as part of you.

5. Export as WAV.

File > Export > Export as WAV, and choose **24-bit PCM**. Not MP3, not M4A. Lossy compression destroys exactly the detail the model needs.

What to send

The WAV file, and a note of any line numbers you re-recorded. That is all. Do not split it, rename it, label it or clean it up. I will cut it into individual takes, build the manifest the training step needs, and flag anything clipped, too quiet, or not matching its transcript.

If something goes wrong

Hissy or distant	Gain too low	You are too far from the mic, or the input gain is too low and the noise floor came up with the signal. Move closer before raising gain.
Boomy or echoing	Room too live	Hang more soft material, or move into a smaller space. A duvet over your head and the mic genuinely works.
Popping on p and b	No pop filter	Angle the mic slightly off-axis from your mouth, or put a sock or foam shield over it.
Clicks and mouth noise	Dry mouth	Room-temperature water. Avoid dairy and coffee beforehand.
Voice changes partway	Fatigue	Stop. Start a fresh file another day and tell me where you stopped.

The script

Read every line, in order. The set is built so that all 40 sounds CHOIR can produce are exercised; skipping lines leaves gaps the model cannot fill later.

Part B: sound coverage

Each line targets particular sounds. Read naturally; do not emphasise the words being tested.

1. The polished birch bridge crossed a shallow stream.
2. Odd thoughts caught her at the office door.
3. That cat sat on a flat black mat.
4. He cut the rough bread and shut the hut door.
5. About a dozen of them arrived from the station.
6. I ought to have bought the tall walnut cabinet.
7. The brown cow found her way out of the crowded house.
8. I tried to hide the bright kite behind the pile.
9. The nurse heard the early bird chirp in the fern.
10. They ate the plain cake they made yesterday.
11. It is a bit thick, and it will fit in this tin.
12. Each evening we eat a clean meal and read a little.
13. He rowed the old boat home alone over cold water.
14. The boy enjoyed the noisy toy his employer destroyed.
15. She took a good look at the wool hood on the hook.
16. Two blue shoes moved through the room to the pool.
17. Ten eggs fell and left a mess on the wet step.
18. Bob grabbed a rubber tube from the cab beside the club.
19. The chief chose cheap cheese and a rich chocolate cherry.
20. Did the dog dig down under the old red shed?
21. They gathered together, breathing in the smooth weather.
22. Four friends found a fifth flag left in the soft office.
23. The big green flag hung against the rugged gate.
24. He held his hat behind him in the hallway.
25. George urged the judge to enjoy the gentle jury.
26. Take the black kite back to the market kiosk.
27. Lily will fill the little yellow bottle slowly.
28. My mother came home and made him a warm meal.

29. Nine noisy children ran down the narrow lane.
30. Singing and ringing, the young king was banging a long gong.
31. Please put the paper cup on the top step.
32. Our friend carried the red barrel around the corner.
33. The sun rose slowly across the misty grass.
34. She should wash the fresh fish in the shallow dish.
35. Take the little metal kettle out to the front gate.
36. I think the thin path through the north is worth it.
37. Every valve moved over the curved silver cover.
38. We waited a while, watching the wet weather worsen.
39. Yesterday the young lawyer used your yellow folder.
40. The lazy zebra dozed in the zoo, buzzing with bees.
41. The casual measure of pleasure was an unusual decision.
42. Who has the whole thing? He said he had it here.
43. A heavy crate of glass bottles arrived this morning.
44. Fasten the loose sheet to the dark wooden frame.
45. The scent of lemons filled the small warm kitchen.

Part C: expression and phrasing

Questions, lists, contrast and pauses. Read these with real but unexaggerated expression, or the voice will be flat.

46. Did you actually read the whole thing, or only the first part?
47. Where are you going?
48. Stop. Put it down and step away from the table.
49. We need bread, milk, two dozen eggs, a bag of flour, and salt.
50. It was not the answer she wanted, but it was the truth.
51. I did not say he stole the money. I said someone did.
52. What a remarkable, extraordinary, entirely unexpected thing to happen.
53. If the weather holds, and if the road is clear, we will leave at dawn.
54. He paused, considered it for a long moment, and then agreed.
55. No. Absolutely not. Not under any circumstances.
56. Wait — did you hear that?
57. She whispered it so quietly that nobody else in the room noticed.
58. The meeting is at half past three on Thursday afternoon.

59. First, we measure. Then we decide. Only then do we build.
60. And that, in the end, is the whole of the matter.

Part D: Scripture and theology

Your subject matter, and the names the engine most needs to say correctly. Read numbers and references exactly as printed.

61. In the beginning God created the heaven and the earth.
62. The book of Ecclesiastes, chapter three, verse one.
63. See First Corinthians thirteen, verses four through seven.
64. Melchizedek, king of Salem, brought forth bread and wine.
65. The letter to the Philippians was written from prison.
66. Habakkuk, Zephaniah, Haggai, Zechariah, and Malachi.
67. Blessed are the peacemakers, for they shall be called the children of God.
68. Read Second Chronicles seven fourteen aloud, slowly.
69. The Nicene Creed speaks of one baptism for the remission of sins.
70. Eschatology, soteriology, and ecclesiology are not the same subject.
71. Deuteronomy six, four. Hear, O Israel.
72. He turned to the Epistle of James and read the first chapter.
73. The Septuagint predates the Masoretic text we use today.
74. Thessalonians, Colossians, Ephesians, and Galatians.
75. Psalm one hundred and nineteen is the longest in the psalter.

When you are done

Send the WAV. The next steps are mine: splitting, training, converting to Core ML, and measuring the result against CHOIR's intelligibility gate. The current rule-based voice scores 8.2 percent word accuracy. That is the number to beat.